
Predictor-Corrector Interior Point Methods

Lecture 4 established what the geometry of the central path allows: the straight line complexity of the max central path lower-bounds the iteration count of every path-following method, and is itself controlled by the circuit imbalance of the constraint matrix. What remains is the algorithm. In this final lecture we construct the classical *predictor-corrector* interior point method and prove its $O(\sqrt{n} \log(1/\varepsilon))$ iteration bound; we then confront its one structural weakness — it cannot terminate finitely, not even on a perfectly straight path — and repair it with the *trust-region step*, obtaining the method whose iteration count matches the straight line complexity, completing the arc of the course.

Throughout, we work with the primal-dual pair

$$\min c^\top x \text{ s.t. } \mathbf{A}x = b, x \geq \mathbf{0}, \quad \max b^\top y \text{ s.t. } \mathbf{A}^\top y + s = c, s \geq \mathbf{0}, \quad (1)$$

with $\mathbf{A} \in \mathbb{R}^{m \times n}$ of full row rank and both programs strictly feasible. We write $z = (x, s, y)$ for a primal-dual point, and recall the gap formula $c^\top x - b^\top y = s^\top x \geq 0$ and complementary slackness: optimal x^*, s^* satisfy $x_i^* s_i^* = 0$ for all i . Products of vectors are coordinatewise: $xs := (x_1 s_1, \dots, x_n s_n)$, and likewise $x/s, \sqrt{x}, x^{-1}$.

1 The Primal-Dual Central Path

Interior point methods solve (1) by relaxing complementary slackness $x_i s_i = 0$ to $x_i s_i = \mu > 0$ and driving $\mu \rightarrow 0$. Consider the primal log-barrier problem of Lecture 4 together with its dual counterpart:

$$x(\mu) := \operatorname{argmin} \left\{ c^\top x - \mu \sum_{i=1}^n \ln x_i : \mathbf{A}x = b, x > \mathbf{0} \right\}, \quad (2)$$

$$(s(\mu), y(\mu)) := \operatorname{argmax} \left\{ b^\top y + \mu \sum_{i=1}^n \ln s_i : \mathbf{A}^\top y + s = c, s > \mathbf{0} \right\}. \quad (3)$$

Both optima exist and are unique for every $\mu > 0$, by strict convexity/concavity and strict feasibility, exactly as in Lecture 4. While not immediately clear, it turns out the both optimizations above are in fact (Lagrangian) duals to each other, in the sense that the value of the primal program upper bounds the value of the dual program after shifting the dual value by an additive $\mu(1 - \ln(\mu))n$. Furthermore, the optimal solutions of these programs are tied together by the following system of equations, which can be derived just from the first-order conditions for optimality.

LEMMA 1 (CENTRAL PATH EQUATIONS) *For every $\mu > 0$, the point $z(\mu) := (x(\mu), s(\mu), y(\mu))$ is uniquely characterized by*

$$\mathbf{A}x = b, x > \mathbf{0}, \quad \mathbf{A}^\top y + s = c, s > \mathbf{0}, \quad xs = \mu \mathbf{1}.$$

PROOF: Let $x := x(\mu)$ be the minimizer of (2): an interior minimizer of the convex function $f(w) := c^\top w - \mu \sum_i \ln w_i$ over the affine subspace $\{w : \mathbf{A}w = b\}$. First-order optimality gives $\nabla f(x) = c - \mu x^{-1} \perp \ker(\mathbf{A})$, and since the orthogonal complement of $\ker(\mathbf{A})$ is $\operatorname{im}(\mathbf{A}^\top)$, there is $y \in \mathbb{R}^m$ with $c - \mu x^{-1} = \mathbf{A}^\top y$. Setting $s := \mu x^{-1} > \mathbf{0}$ then yields $\mathbf{A}^\top y + s = c$ and $xs = \mu \mathbf{1}$: the

triple (x, s, y) solves the displayed system. Conversely, for any solution (x', s', y') of the system, x' is strictly feasible with $\nabla f(x') = c - \mu(x')^{-1} = \mathbf{A}^\top y' \perp \ker(\mathbf{A})$, so x' is a minimizer of f ; by uniqueness of the minimizer, $x' = x(\mu)$, and then $s' = \mu(x')^{-1}$ and y' are determined as well (using that $\mathbf{A}$ has full row rank).

The same argument applied to (3) — substitute $s = c - \mathbf{A}^\top y$ and set the gradient in y to zero — shows that its optimality condition is $\mathbf{A}(\mu s^{-1}) = b$, i.e., the *same* system with $x := \mu s^{-1}$. Hence the maximizer of (3) is exactly $(s(\mu), y(\mu))$ from the triple above. $\square$

The curve $\mu \mapsto z(\mu)$ is the *primal-dual central path*; its primal component is the central path of Lecture 4. Summing the centrality equation, the optimality gap on the path is

$$c^\top x(\mu) - b^\top y(\mu) = x(\mu)^\top s(\mu) = n\mu,$$

so following the path to $\mu \rightarrow 0$ solves both programs. The limit $z^* := \lim_{\mu \rightarrow 0} z(\mu)$ exists and is a *strictly complementary* optimal pair: for each i , exactly one of $x_i^* > 0$, $s_i^* > 0$ holds, inducing the optimality partition $B \cup N = [n]$ with $B = \text{supp}(x^*)$, $N = \text{supp}(s^*)$ (we take this classical fact on faith; it is due to Goldman and Tucker).

Exercise 1 (Path rescaling) Let $z(1) = (x(1), s(1), y(1))$ be the central path point at $\mu = 1$. Show that rescaling the variables by $x_j \leftarrow x_j/x_j(1)$ (and correspondingly $\mathbf{A}_j \leftarrow x_j(1)\mathbf{A}_j$, $c_j \leftarrow x_j(1)c_j$, $s_j \leftarrow s_j/s_j(1)$) maps the instance to an equivalent one whose central path point at $\mu = 1$ is $(x, s) = (\mathbf{1}, \mathbf{1})$, and maps central path to central path. (This normalization is used in Section 3.)

2 The Predictor-Corrector Method

We now build a method that tracks the central path. The iterates will be strictly feasible points close to the path in the following multiplicative sense.

DEFINITION 2 (NORMALIZED GAP, NEIGHBORHOOD) For strictly feasible $z = (x, s, y)$, the normalized gap is $\mu(z) := x^\top s/n$. For $\beta \in (0, 1)$, the ℓ_2 -neighborhood of the central path is

$$N_2(\beta) := \{z : \left\| \frac{xs}{\mu(z)} - \mathbf{1} \right\|_2 \leq \beta\},$$

and $\overline{N_2}(\beta)$ denotes its closure (which contains the limit z^*).

On the path, $xs = \mu\mathbf{1}$; in $N_2(\beta)$, every product $x_i s_i$ lies within a factor $(1 \pm \beta)$ of the average $\mu(z)$, since the ℓ_∞ norm is bounded by the ℓ_2 norm:

$$z \in N_2(\beta) \implies (1 - \beta)\mu(z) \leq x_i s_i \leq (1 + \beta)\mu(z) \quad \forall i \in [n]. \quad (4)$$

The method, due to Mizuno, Todd and Ye [MTY93] (building on the primal-dual framework of Kojima, Mizuno and Yoshise [KMY89]), alternates two kinds of Newton steps. Both arise from linearizing a target equation: we seek $\Delta z = (\Delta x, \Delta s, \Delta y)$ with

$$s \Delta x + x \Delta s = t, \quad \mathbf{A} \Delta x = \mathbf{0}, \quad \mathbf{A}^\top \Delta y + \Delta s = \mathbf{0}, \quad (5)$$

where the *target* $t \in \mathbb{R}^n$ is the linearization of what we want $(x + \Delta x)(s + \Delta s)$ to become: dropping the quadratic term $\Delta x \Delta s$ from

$$(x + \Delta x)(s + \Delta s) = xs + s \Delta x + x \Delta s + \Delta x \Delta s,$$

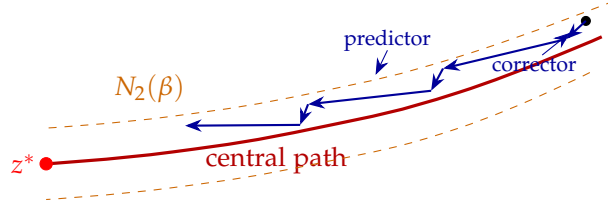


Figure 1: The predictor-corrector method: correctors pull the iterate back towards the path at constant gap; predictors move down the tube, shrinking the gap by the factor $(1 - \alpha)$.

the choice $t = \mu(z)\mathbf{1} - xs$ aims at the central path point with the current gap (*corrector step*, target $xs \rightarrow \mu(z)\mathbf{1}$), while $t = -xs$ aims directly at the optimum (*predictor step*, target $xs \rightarrow \mathbf{0}$; this direction is also known as *affine scaling*). In both cases the whole error of the step is precisely the neglected quadratic term $\Delta x \Delta s$ — an observation to remember for Section 3.

The steps as projections. The last two equations of (5) say $\Delta x \in \ker(\mathbf{A})$ and $\Delta s \in \text{im}(\mathbf{A}^\top) = \ker(\mathbf{A})^\perp$; in particular $\Delta x^\top \Delta s = 0$ for every step of this form. Multiplying the first equation by $(xs)^{-1/2}$ gives

$$\underbrace{\sqrt{\frac{s}{x}} \Delta x}_{\in \ker(\mathbf{A}D)} + \underbrace{\sqrt{\frac{x}{s}} \Delta s}_{\in \text{im}((\mathbf{A}D)^\top)} = \frac{t}{\sqrt{xs}}, \quad D := \text{diag}(\sqrt{x/s}),$$

an *orthogonal decomposition* of the vector $t/\sqrt{xs}$ into complementary subspaces. Hence (5) always has a unique solution (computable by one linear system solve), and Pythagoras yields the identity we will use repeatedly:

$$\left\| \sqrt{\frac{s}{x}} \Delta x \right\|_2^2 + \left\| \sqrt{\frac{x}{s}} \Delta s \right\|_2^2 = \left\| \frac{t}{\sqrt{xs}} \right\|_2^2. \quad (6)$$

Algorithm 1: PREDICTOR-CORRECTOR IPM [MTY93]

Input: $z = (x, s, y) \in N_2(\beta)$, $\beta \in (0, 1/2]$, target gap $\varepsilon \geq 0$.

- 1 **subroutine** DIRECTION(point $z = (x, s, y)$, target $t \in \mathbb{R}^n$):
- 2 **return** the unique solution $\Delta z = (\Delta x, \Delta s, \Delta y)$ to
- 3 $s \Delta x + x \Delta s = t$, $\mathbf{A} \Delta x = \mathbf{0}$, $\mathbf{A}^\top \Delta y + \Delta s = \mathbf{0}$
- 4 **while** $x^\top s > \varepsilon$ **do**
- 5 corrector direction: $\Delta z^c \leftarrow \text{DIRECTION}(z, \mu(z)\mathbf{1} - xs)$;
- 6 corrector step: $z \leftarrow z + \Delta z^c$;
- 7 predictor direction: $\Delta z^p \leftarrow \text{DIRECTION}(z, -xs)$;
- 8 predictor step: take the largest $\alpha \in [0, 1]$ with $z + \alpha' \Delta z^p \in \overline{N_2}(\beta)$ for all $\alpha' \in [0, \alpha]$;
- 9 $z \leftarrow z + \alpha \Delta z^p$;
- 9 **return** z

2.1 Analysis

Two elementary facts do all the work. First, for any $u, v \in \mathbb{R}^n$,

$$\|uv\|_2 \leq \|u\|_2 \|v\|_2 \leq \frac{1}{2}(\|u\|_2^2 + \|v\|_2^2), \quad (7)$$

(the first inequality holds since $\sum_i u_i^2 v_i^2 \leq (\sum_i u_i^2)(\sum_j v_j^2)$, the second by AM–GM). Second, both steps preserve or scale the gap exactly: since $\Delta x^\top \Delta s = 0$,

$$n \mu(z + \alpha \Delta z) = (x + \alpha \Delta x)^\top (s + \alpha \Delta s) = x^\top s + \alpha \mathbf{1}^\top t, \quad (8)$$

which equals $x^\top s$ for the corrector ($\mathbf{1}^\top t = n\mu(z) - x^\top s = 0$) and $(1 - \alpha) x^\top s$ for the predictor ($\mathbf{1}^\top t = -x^\top s$).

LEMMA 3 (CORRECTOR) *Let $z \in N_2(\beta)$, $\beta \in (0, 1/2]$. Then $\mu(z + \Delta z^c) = \mu(z)$ and $z + \Delta z^c \in N_2(\frac{\beta^2}{2(1-\beta)}) \subseteq N_2(\beta/2)$.*

PROOF: The gap claim is (8). For membership in N_2 , write $\mu := \mu(z)$; by the target equation, $(x + \Delta x^c)(s + \Delta s^c) = \mu \mathbf{1} + \Delta x^c \Delta s^c$, and the gap of $z + \Delta z^c$ is again μ , so

$$\left\| \frac{(x + \Delta x^c)(s + \Delta s^c)}{\mu} - \mathbf{1} \right\|_2 = \left\| \frac{\Delta x^c \Delta s^c}{\mu} \right\|_2 = \left\| \left(\sqrt{\frac{s}{x\mu}} \Delta x^c \right) \left(\sqrt{\frac{x}{s\mu}} \Delta s^c \right) \right\|_2 \stackrel{(7),(6)}{\leq} \frac{1}{2} \left\| \frac{\mu \mathbf{1} - xs}{\sqrt{\mu xs}} \right\|_2^2.$$

By (4), each denominator entry satisfies $x_i s_i \geq (1 - \beta)\mu$, so the right-hand side is at most $\frac{1}{2(1-\beta)} \left\| \frac{\mu \mathbf{1} - xs}{\mu} \right\|_2^2 \leq \frac{\beta^2}{2(1-\beta)}$. Finally $\frac{\beta^2}{2(1-\beta)} \leq \beta^2 \leq \beta/2$ for $\beta \leq 1/2$. (Strict positivity of $x + \Delta x^c$, $s + \Delta s^c$ follows since the products stay positive along the segment.) $\square$

LEMMA 4 (PREDICTOR) *Let $z \in N_2(\beta/2)$, $\beta \in (0, 1/2]$. Then:*

1. $\mu(z + \alpha \Delta z^p) = (1 - \alpha) \mu(z)$ for all α ;
2. the normalized quadratic error of the predictor direction satisfies $\|\Delta x^p \Delta s^p / \mu(z)\|_2 \leq n/2$;
3. every $\alpha \in [0, 1)$ with $\frac{\alpha^2}{1-\alpha} \|\Delta x^p \Delta s^p / \mu(z)\|_2 \leq \frac{\beta}{2}$ keeps $z + \alpha \Delta z^p \in N_2(\beta)$; in particular the step size in Algorithm 1 satisfies $\alpha \geq \frac{1}{2} \sqrt{\beta/n}$.

PROOF: (1) is (8). For (2), exactly as in the corrector proof (writing $\mu := \mu(z)$),

$$\left\| \frac{\Delta x^p \Delta s^p}{\mu} \right\|_2 \stackrel{(7),(6)}{\leq} \frac{1}{2} \left\| \frac{-xs}{\sqrt{\mu xs}} \right\|_2^2 = \frac{1}{2} \sum_{i=1}^n \frac{x_i s_i}{\mu} = \frac{n}{2}.$$

For (3), using the target equation, $(x + \alpha \Delta x^p)(s + \alpha \Delta s^p) = (1 - \alpha)xs + \alpha^2 \Delta x^p \Delta s^p$, and $\mu(z + \alpha \Delta z^p) = (1 - \alpha)\mu$, so by the triangle inequality

$$\left\| \frac{(x + \alpha \Delta x^p)(s + \alpha \Delta s^p)}{(1 - \alpha)\mu} - \mathbf{1} \right\|_2 = \left\| \frac{xs}{\mu} - \mathbf{1} + \frac{\alpha^2}{1 - \alpha} \frac{\Delta x^p \Delta s^p}{\mu} \right\|_2 \leq \frac{\beta}{2} + \frac{\alpha^2}{1 - \alpha} \left\| \frac{\Delta x^p \Delta s^p}{\mu} \right\|_2.$$

The condition in (3) keeps this at most β . Finally, $\alpha := \frac{1}{2} \sqrt{\beta/n}$ satisfies it: then $\alpha \leq \frac{1}{2\sqrt{2}} < \frac{1}{2}$, so by (2),

$$\frac{\alpha^2}{1 - \alpha} \left\| \frac{\Delta x^p \Delta s^p}{\mu} \right\|_2 \leq 2\alpha^2 \cdot \frac{n}{2} = \frac{\beta}{4} \leq \frac{\beta}{2}.$$

$\square$

THEOREM 5 (ITERATION BOUND [MTY93]) From $z_0 \in N_2(\beta)$, $\beta \in (0, 1/2]$, Algorithm 1 maintains $z \in \bar{N}_2(\beta)$ with

$$x^\top s \leq (x_0^\top s_0) e^{-\frac{1}{2}\sqrt{\beta/n} T} \quad \text{after } T \text{ iterations,}$$

and hence reaches gap $\varepsilon > 0$ within $2\sqrt{n/\beta} \ln(x_0^\top s_0/\varepsilon) + 1$ iterations.

PROOF: By induction: from $z \in N_2(\beta)$, the corrector yields a point of the same gap in $N_2(\beta/2)$ (Theorem 3), from which the predictor multiplies the gap by $(1 - \alpha) \leq 1 - \frac{1}{2}\sqrt{\beta/n} \leq e^{-\frac{1}{2}\sqrt{\beta/n}}$ while returning to $\bar{N}_2(\beta)$ (Theorem 4). The iteration count follows by taking logarithms. $\square$

Every iteration costs $O(1)$ linear system solves. With β constant, the gap halves every $O(\sqrt{n})$ iterations — Renegar’s bound [Ren88], now with a self-contained proof — and over the rationals, running until $\varepsilon = 2^{-\Theta(L)}$ and rounding yields an exact solution in $O(\sqrt{n} L)$ iterations: the weakly polynomial bound of the Lecture 4 landscape.

Initialization. Algorithm 1 needs a strictly feasible, well-centered starting point, which (1) need not possess — the LP may not even be feasible. The remedy of Ye, Todd and Mizuno [YTM94] is to embed (1) into a larger LP that is feasible *by design*. Two new scalar variables do the work: a homogenizing variable $\tau \geq 0$ that scales the data, and a variable θ that *relaxes feasibility*, weighted by the residual vectors of the intended starting point $(x, s, y) = (\mathbf{1}, \mathbf{1}, \mathbf{0})$:

$$\bar{b} := b - \mathbf{A}\mathbf{1}, \quad \bar{c} := c - \mathbf{1}, \quad \bar{g} := c^\top \mathbf{1} + 1.$$

The *homogeneous self-dual embedding* is the LP

$$\begin{aligned} \min \quad & (n+1)\theta \\ \text{s.t.} \quad & \mathbf{A}x - b\tau + \bar{b}\theta = \mathbf{0} \\ & -\mathbf{A}^\top y + c\tau - \bar{c}\theta = s, \quad s \geq \mathbf{0} \\ & b^\top y - c^\top x + \bar{g}\theta = \kappa, \quad \kappa \geq 0 \\ & -\bar{b}^\top y + \bar{c}^\top x - \bar{g}\tau = -(n+1), \end{aligned} \tag{9}$$

with $x \geq \mathbf{0}$, $\tau \geq 0$ and y, θ free. The first three rows are the combined feasibility and zero-gap conditions of (1) and its dual, scaled by τ and *relaxed* by θ ; the fourth row normalizes θ . By construction, $\theta = 1$ absorbs exactly the residuals of the all-ones point:

$$(y, x, \tau, \theta) = (\mathbf{0}, \mathbf{1}, 1, 1) \quad \text{is feasible, with slacks } (s, \kappa) = (\mathbf{1}, 1),$$

and this point is *perfectly centered*: every product $x_i s_i = \tau \kappa = 1$. Moreover, the coefficient matrix on the left of (9) is *skew-symmetric*, which makes the LP its own dual and yields, for every feasible point, the identity

$$x^\top s + \tau \kappa = (n+1)\theta \tag{10}$$

(Exercise 2(a)). The objective is therefore precisely the complementarity gap of the embedding: it is nonnegative on the feasible region, and the all-ones point is a strictly feasible, perfectly centered primal-dual pair, so the central path of (9) exists and Algorithm 1 applies verbatim — with (x, τ) in the primal role, (s, κ) in the dual role, and (y, θ) as multipliers — driving $(n+1)\theta = x^\top s + \tau \kappa \rightarrow 0$. The strictly complementary limit (Section 1) has $\theta = 0$, $x^\top s = \tau \kappa = 0$, and exactly one of $\tau > 0$, $\kappa > 0$. If $\tau > 0$, then $(x/\tau, s/\tau, y/\tau)$ is an optimal primal-dual pair for (1); if $\kappa > 0$, the point furnishes a Farkas certificate that (1) or its dual is infeasible (Exercise 2). Either way, the original LP is decided with no feasibility assumption at all.

Exercise 2 Verify the claims about the embedding (9): (a) check that the all-ones point is feasible, and derive (10) by multiplying the four constraint groups by y, x, τ, θ respectively and adding — all data terms cancel in skew-symmetric pairs; (b) show that if an optimal solution has $\tau > 0$, then $(x/\tau, s/\tau, y/\tau)$ is an optimal primal-dual pair for (1); (c) show that if instead $\kappa > 0$, then $\tau = 0$ and either $c^\top x < 0$, in which case x certifies dual infeasibility via $\mathbf{A}x = \mathbf{0}, x \geq \mathbf{0}, c^\top x < 0$, or $b^\top y > 0$, in which case y certifies primal infeasibility via $\mathbf{A}^\top y \leq \mathbf{0}, b^\top y > 0$.

3 Reaching the End of the Path: the Trust-Region Step

Theorem 5 has a structural blind spot: the algorithm *never finishes*. The gap decreases geometrically but remains positive forever; exact termination is obtained only by the exogenous rounding step, whose threshold $2^{-\Theta(L)}$ depends on the bit size of the data — precisely what a strongly polynomial algorithm is forbidden from using. If interior point methods are ever to match the clean, finite termination of the simplex method — and, via the circuit cover theorem of Lecture 4, to achieve strongly polynomial iteration bounds when κ_A is bounded — the method itself must be able to *reach the end of the path*.

What makes this frustrating is that the end of the path is the *easiest possible geometry*: it is asymptotically a straight line. We make this precise, then show that the predictor step is structurally unable to walk it, and finally present the step that can.

3.1 The last stretch of the path is straight — and polarized

Suppose the path has essentially converged in shape: after the rescaling of Exercise 1 we sit at $z(1)$ with $(x(1), s(1)) = (\mathbf{1}, \mathbf{1})$, and we assume the current point is close to the limit $z^* = (x^*, s^*, y^*)$ in the sense

$$\|(x_B^* - \mathbf{1}_B, s_N^* - \mathbf{1}_N)\|_2 \leq \beta, \quad (11)$$

where $B \cup N$ is the optimality partition ($x_N^* = \mathbf{0}, s_B^* = \mathbf{0}$). Consider the *one step to the end*: $\Delta z := z^* - z(1)$. Its shape is the

$$\text{polarization pattern: } \underbrace{(\Delta x_B, \Delta x_N)}_{\|\cdot\| \leq \beta} \approx (\mathbf{0}_B, -\mathbf{1}_N), \quad \underbrace{(\Delta s_B, \Delta s_N)}_{\|\cdot\| \leq \beta} \approx (-\mathbf{1}_B, \mathbf{0}_N) : \quad (12)$$

the primal step wipes out the N -coordinates uniformly while barely moving the B -coordinates, and the dual step does the mirror image. The whole segment to the optimum lies in the neighborhood:

PROPOSITION 6 (THE FINAL SEGMENT IS CENTRAL) *Under (11), the segment $z(1) + \alpha \Delta z$, $\alpha \in [0, 1]$, satisfies, for all α ,*

$$(\mathbf{1} + \alpha \Delta x)(\mathbf{1} + \alpha \Delta s) = (1 - \alpha)(\mathbf{1} + \alpha e), \quad e := (\Delta x_B, \Delta s_N), \quad \|e\|_2 \leq \beta,$$

so that, after normalizing the gap by $(1 - \alpha)$, the products deviate from $\mathbf{1}$ by αe , of ℓ_2 norm at most $\alpha\beta \leq \beta$: the ideal step travels the entire remaining path inside (a negligible enlargement of) $N_2(\beta)$, reaching the exact optimum at $\alpha = 1$.

PROOF: Compute the products separately on the two blocks. On N : $\Delta x_N = -\mathbf{1}_N$, so $(1 - \alpha)(1 + \alpha \Delta s_N)$ exactly. On B : $\Delta s_B = -\mathbf{1}_B$, so $(1 - \alpha)(1 + \alpha \Delta x_B)$ exactly. Both are of the claimed

form, and $\|(\Delta x_B, \Delta s_N)\|_2 \leq \beta$ by (11). Dividing by the gap factor $(1 - \alpha)$ and subtracting $\mathbf{1}$ leaves αe , of norm at most $\alpha\beta$. (The exact normalized gap differs from $(1 - \alpha)$ by the factor $1 + \alpha \mathbf{1}^T e/n = 1 + O(\beta/\sqrt{n})$, which we suppress.) $\square$

3.2 Why the predictor cannot take this step

The predictor direction satisfies the affine scaling equation $s\Delta x + x\Delta s = -xs$, which at $(x, s) = (\mathbf{1}, \mathbf{1})$ reads $\Delta x + \Delta s = -\mathbf{1}$: *coordinatewise*, the primal and dual steps split the decrease -1 between them. Now look at what the polarized step (12) demands on a coordinate $i \in B$: $(\Delta x_i, \Delta s_i) \approx (0, -1)$, so that the product becomes $(1 + \Delta x_i)(1 + \Delta s_i) \approx 0$ with error $|\Delta x_i \Delta s_i| \approx |\Delta x_i|$ — small precisely because Δx_i is small. The affine scaling step can never produce this shape. Consider a coordinate that must decrease its product to $\nu \ll 1$ while retaining a deviation of size β on the s -side — the pattern $(\Delta x_i, \Delta s_i) = (-1 + \nu, \beta)$. Its product is $\nu(1 + \beta)$: an error of only $\nu\beta$, exactly what the neighborhood $N_2(\beta)$ affords at gap ν . But its quadratic term is $|\Delta x_i \Delta s_i| \approx \beta \gg \nu\beta$, and the error committed by the affine scaling step *equals* its quadratic term — so this pattern is out of its reach. In short: *the predictor distributes its decrease between x and s on every coordinate as its projection equations dictate, while long steps require all the decrease on one side, with the side depending on the coordinate*. The predictor's step length is capped by its quadratic error — which generically does not vanish even when the path ahead is perfectly straight — so that the gap decreases by at most a bounded factor per step, and the method never terminates finitely (Exercise 3).

This diagnosis also explains what long steps look like anywhere on the path, not just at the end. Call the piece of the central path between parameters $\mu_1 < \mu_0$ *polarized* if there is a partition $B' \cup N' = [n]$ such that, on the whole piece, $x_{B'}(\mu) \approx x_{B'}(\mu_0)$ and $x_{N'}(\mu) \approx (\mu/\mu_0)x_{N'}(\mu_0)$, with the roles swapped for s : each coordinate either *stays put* or *scales linearly with the gap* (Figure 2). These are exactly the straight stretches of the path on a multiplicative scale — and exactly the segments of the piecewise linear approximations behind the straight line complexity of Lecture 4, where each linear piece was, up to a factor 2, either constant or a scaling (Exercise 4).

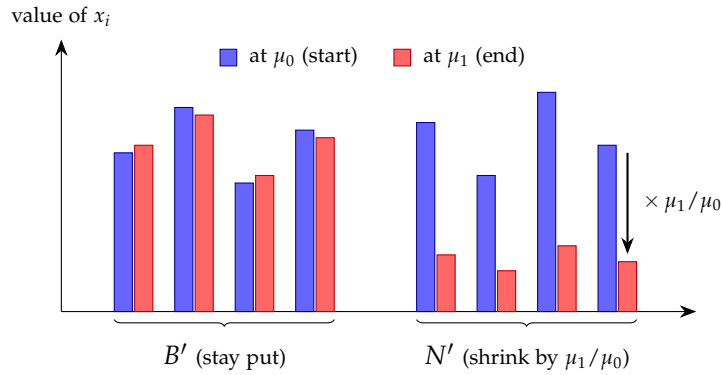


Figure 2: The coordinates of x across a polarized segment of the central path, traversed from gap μ_0 (blue bars) down to μ_1 (red bars): each B' -coordinate (left) stays roughly the same, while the N' -coordinates (right) all shrink by the *same* factor μ_1/μ_0 . For s the roles of B' and N' are swapped. The trust-region step traverses such a segment in a single step.

Exercise 3 (Exact termination requires polarization) (a) Show that a predictor step with step size $\alpha = 1$ lands on a feasible optimal pair if and only if $\Delta x^p \Delta s^p = \mathbf{0}$ coordinatewise — i.e.,

the affine scaling direction is already *perfectly polarized*: on each coordinate, either $\Delta x_i^p = -x_i$ and $\Delta s_i^p = 0$, or the reverse. (Hint: Compute the products at $\alpha = 1$.) (b) Verify that the perfectly polarized pattern is consistent with the coordinatewise equation $\Delta x_i/x_i + \Delta s_i/s_i = -1$, and explain why the subspace constraints $\Delta x \in \ker(\mathbf{A})$, $\Delta s \in \text{im}(\mathbf{A}^\top)$ — which force Δz^p to be the *orthogonal* decomposition of Section 2 — generically rule it out. (The trust-region step recovers the polarized pattern by imposing it through the constraint sets instead.)

3.3 The trust-region step

The step that traverses a polarized segment must be told the partition and asked for the right asymmetric motion: shrink $x_{N'}$ as much as possible while disturbing $x_{B'}$ as little as possible (and dually). The cleanest formalization, due to Lan, Monteiro and Tsuchiya [LMT09], is a pair of *trust region* problems:

$$\Delta x^{\text{tr}} := \operatorname{argmin} \left\{ \left\| \frac{x_{N'} + \Delta x_{N'}}{x_{N'}} \right\|_2 : \mathbf{A} \Delta x = \mathbf{0}, \left\| \frac{\Delta x_{B'}}{x_{B'}} \right\|_2 \leq \frac{1}{10} \right\}, \quad (13)$$

$$\Delta s^{\text{tr}} := \operatorname{argmin} \left\{ \left\| \frac{s_{B'} + \Delta s_{B'}}{s_{B'}} \right\|_2 : \Delta s \in \text{im}(\mathbf{A}^\top), \left\| \frac{\Delta s_{N'}}{s_{N'}} \right\|_2 \leq \frac{1}{10} \right\} : \quad (14)$$

minimize the relative residual of the coordinates that should vanish, subject to a *trust region* constraint protecting the coordinates that should stay put. (Each problem minimizes a convex quadratic over a Euclidean ball intersected with a subspace, and is solvable with a few linear system solves.) The guarantee, which we state without proof, is that the step is as long as the geometry allows:

LEMMA 7 (TRUST-REGION GUARANTEE [LMT09, ADL⁺22]) *Let $z \in N_2(\beta/2)$, for $\beta \in (0, 1/6]$, and let*

$$r := \max \left\{ \left\| \frac{x_{N'} + \Delta x_{N'}^{\text{tr}}}{x_{N'}} \right\|_2, \left\| \frac{s_{B'} + \Delta s_{B'}^{\text{tr}}}{s_{B'}} \right\|_2 \right\}$$

denote the attained residual. If $r \leq \beta/20$, then the combined step satisfies

$$z + (1 - 10r/\beta) \Delta z^{\text{tr}} \in N_2(\beta), \quad \mu(z + (1 - 10r/\beta) \Delta z^{\text{tr}}) = O(r) \mu(z).$$

On a polarized segment reaching down to parameter μ_1 , the attained residual is $r = O(\mu_1/\mu_0)$, so a *single* trust-region step (followed by correctors) traverses the whole segment; at the end of the path, where the residual can reach $r = 0$, it lands exactly on z^* — finite termination with no rounding.

One ingredient is missing: the partition. Remarkably, the predictor direction — the very step that cannot walk the segment — *reveals* it: since affine scaling splits -1 between $\Delta x_i/x_i$ and $\Delta s_i/s_i$ on each coordinate, on a polarized segment the split is lopsided in the direction of the polarization, and one simply reads off

$$N' := \left\{ i : \frac{\Delta x_i^p}{x_i} < \frac{\Delta s_i^p}{s_i} \right\}, \quad B' := [n] \setminus N'. \quad (15)$$

Algorithm 2: TRUST-REGION IPM [LMT09]

Input: $z \in N_2(\beta)$, β a small constant.

```
1 while  $x^\top s > 0$  do
2   corrector step at  $z$ ;
3   predictor step: compute  $\Delta z^p$  and maximal  $\alpha$ ;
4   if  $\alpha \geq 3/4$  then
5     guess the partition  $(B', N')$  from  $\Delta z^p$  via (15);
6     solve (13)–(14); take the trust-region step of Theorem 7 if it improves on  $z + \alpha \Delta z^p$ ,
       else the predictor step;
7   else
8     take the predictor step  $z \leftarrow z + \alpha \Delta z^p$ ;
9 return  $z$ 
```

THEOREM 8 (ITERATIONS MATCH STRAIGHT LINE COMPLEXITY [ADL⁺22]) *Started at a well-centered z_0 with gap g_0 , the TRUST-REGION IPM computes an exact optimal pair $z^* \in \overline{N}_2(\beta)$ within*

$$O\left(\min_{\gamma \in (0,1]} n^5 \log\left(\frac{n}{\gamma}\right) \sum_{i=1}^n \text{SLC}_\gamma(x_i^m, g_0)\right) \quad \text{iterations.}$$

The proof, which we do not present, runs along the lines this lecture has set up: decompose the (max) central path into $\sum_i \text{SLC}_\gamma(x_i^m)$ approximately polarized segments; show that on each segment the partition guess (15) is correct once the segment is entered; and show that the trust-region step then traverses the segment in $O(n^5 \log(n/\gamma))$ predictor-corrector-and-trust-region iterations, in an amortized sense. The $\sqrt{n}$ -type factor is inherited from Theorem 4, and the sum over the SLCs is what bounds the number of polarized segments of the full primal-dual path.

Let us now highlight what Theorem 8 achieves in relation to previous lectures.

Exact termination. The Trust-Region IPM terminates at an optimal solution after a finite number of iterations by following the central path until the end. That is, it turns what is classically an approximate method into an exact one, just like the simplex method.

Matching the lower bound. By Lecture 4, no path-following method can beat $\max_i \text{SLC}(x_i^m)$ iterations; Theorem 8 achieves $\tilde{O}(n^5) \sum_i \text{SLC}(x_i^m)$. Up to this polynomial factor, path-following interior point methods are *exactly as fast as the geometry of the central path allows* — and the tropical instances of Lecture 4 provide examples where this geometry itself can be exponential.

LPs with strongly polynomial SLC. Combining with the circuit cover bound of Lecture 4, namely $\text{SLC}_{1/(n\kappa_A)^2}(x_i^m) \leq \min\{m, n - m\} + 2$, the Trust-Region IPM runs in

$$O(n^{1.5} \min\{m, n - m\} \log(n\kappa_A))$$

iterations — strongly polynomial whenever κ_A is bounded, recovering the Vavasis–Ye and Tardos-type results of the Lecture 2 landscape, and covering in particular all network matrices ($\kappa = 1$): the min-cost flow problems of Lecture 3 admit strongly polynomial interior point algorithms. [DKN⁺24] further showed that this method can be used to solve any LP where A has at most two non-zero entries per column in strongly polynomial time.

Literature on the Trust-Region IPM. The Trust-Region IPM and the trust-region problems (13)–(14) were developed by Lan, Monteiro and Tsuchiya [LMT09]. Theorem 8 is implicit in [ADL⁺22]:

the algorithm analyzed there follows the same scheme but works with a very approximate version of the trust-region step, which suffices for the straight-line-complexity bound. Dadush, Ma, Natura and Vég  [DMNV26] gave an efficient — strongly polynomial — implementation of the trust-region step itself, and showed that the Trust-Region IPM is moreover *optimal for ℓ_2 path following*: its iteration count is within a constant factor of that of any algorithm whose iterates stay in the same ℓ_2 neighborhood of the central path.

Exercise 4 Recall from Lecture 4 that $\text{SLC}_\gamma(x_i^m, g_0)$ is witnessed by a piecewise linear h with $\gamma x_i^m \leq h \leq x_i^m$. Prove that each linear piece of a minimal h satisfies that $h(g) = ag + b$ where $a, b \geq 0$. Conclude that up to a factor 2, each linear piece is either a *constant* or a *linear scaling* of the gap. Finally show that an SLC decomposition of all coordinates induces a partition of the path into $O(\sum_i \text{SLC}_\gamma(x_i^m))$ segments on which every coordinate either stays put or scales — the polarized segments of this lecture.

3.4 Open Problems and Further Reading

The main open problem remains: is there a strongly polynomial algorithm for all of linear programming? While the full problem cannot be solved via the standard path-following paradigm, due to the exponential lower bound, it is an exciting open question to gain a more complete understanding of which classes of linear programs have strongly polynomial straight line complexity.

In the context of polyhedral diameter, one question is whether one can construct super quadratic lower bounds for connected layer families (CLFs), which is a purely combinatorial problem.

Another important question is to understand why the simplex method is so effective at reoptimization: where the simplex method really shines is in solving the sequence of LPs that arise in branch & bound, where at each stage one re-solves the LP after an additional variable is fixed to zero or one. In this context, the simplex method on the dual receives a new variable each time a constraint in the primal is fixed, and so it always maintains a feasible solution from which it can reoptimize. These reoptimization steps tend to require far fewer pivots than solving the original problem, a behavior that so far lacks any theoretical explanation.

Further reading. The predictor-corrector method and its analysis follow Mizuno, Todd and Ye [MTY93]; the self-dual initialization is due to Ye, Todd and Mizuno [YTM94]. The trust-region step originates with Lan, Monteiro and Tsuchiya [LMT09], and the straight line complexity framework, the matching upper bound, and its consequences are developed in [ADL⁺22, DKN⁺24]; the strongly polynomial trust-region step and the ℓ_2 -optimality of the Trust-Region IPM are from [DMNV26]; see [ENV22] for a survey of the circuit imbalance measure.

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